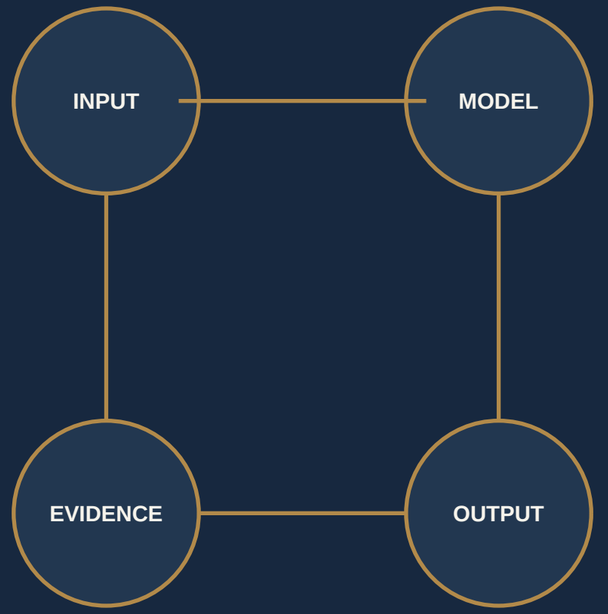
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Digital Signal Processing

AI.108 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define discrete signals.
- Apply key methods including sampling.
- Build or analyze a practical system involving quantization.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

Discrete signals

02

METHODS

Sampling

03

SYSTEMS

Quantization

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · Discrete signals

Discrete signals

02 · Sampling

Sampling

03 · Quantization

Quantization

04 · Convolution

Convolution

05 · Filtering

Filtering

06 · Fourier analysis

Fourier analysis

07 · Frequency-domain analysis ar

Frequency-domain analysis and digital filters

08 · Waveform animations and freque

Waveform animations and frequency-spectrum visualizations

Core ideas, made explicit

AI.108 • Distinguish definitions, assumptions and evidence

- Discrete signals
- Sampling
- Quantization
- Convolution
- Filtering

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.108 • Distinguish definitions, assumptions and evidence

- Fourier analysis
- Frequency-domain analysis and digital filters
- Waveform animations and frequency-spectrum visualizations
- Mandatory practicals: Capture
- Process and analyze real signals using Python/MATLAB or equivalent tools

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

PROBLEM
Discrete signals

INPUTS
Sampling

METHOD
Quantization

OUTPUT
Convolution

EVALUATION
Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving discrete signals.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using discrete signals.
- Implement or simulate a baseline using sampling.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.